11+ Non-Verbal Reasoning

Non-Verbal Reasoning Preparation Guide

Series 1: The Complete Roadmap

WHAT THIS GUIDE COVERS

A complete breakdown of every NVR topic tested at 11+, the SNAPSS method for finding hidden rules, a step-by-step guide to 3D cube questions, two fully worked walkthroughs, and a structured revision routine.

PREPARATION SERIES 1 · SHARE WITH FAMILIES Non-Verbal Reasoning · 11+

ID	Section	Priority
01	Why NVR is the Differentiator	High
02	The Complete 11+ NVR Topic List	Essential
03	Mnemonics and Techniques	Critical
04	Worked Question Walkthroughs	Practical
05	How to Structure Your Revision	Strategy

> *"NVR rewards the student who looks **systematically,** not the one who looks the hardest."*

01 Non-Verbal Reasoning

Why NVR is the Differentiator

4 min read · Section 01 of 05

NVR tests something different from every other 11+ subject. It does not reward revision of facts or formulas. It rewards a specific habit of observation that must be trained deliberately.

A. WHAT NVR ACTUALLY MEASURES

Non-verbal reasoning evaluates a child's ability to identify patterns, manage mental transformations, and solve problems using visual logic. It is not about drawing or art. It is about seeing structure.

Unlike Maths, which responds to short-term drilling, NVR measures a habit of visual discipline. A student who looks at a shape and thinks "that looks right" will consistently lose marks to a student who looks and thinks "I need to check the size, the shading, the angle, and the position before I choose."

B. WHAT EXAMINERS DO TO CATCH STUDENTS OUT

The 11+ paper regularly uses Psychometric Distractors: answer options that are designed to look almost correct. The distractor will match the target on three or four attributes but fail on one subtle detail, such as a slightly different rotation, a stripe going the wrong way, or one extra dot.

The only defence is Active Observation: scanning for specific attributes one at a time, using a checklist, rather than looking at the shape as a whole and relying on instinct.

| The core habit to build

- Never look at an answer option until you have predicted what the answer should look like.
- Always check every attribute of your chosen answer before committing.
- When two options look identical, the difference is always there. Look harder at the detail you have not checked yet.

The one mindset shift that changes everything

Stop asking: "Does this look right?" Start asking: "Can I prove this is correct?" These are different questions. The first relies on intuition. The second relies on a systematic check. NVR is won by the second approach every time.

02 Non-Verbal Reasoning

The Complete 11+ NVR Topic List

6 min read · Section 02 of 05

Every topic tested across GL Assessment, CEM Select, ISEB, and Quest Admissions. These twelve topics cover the full scope of what your child will encounter on the day.

INTERPRETING SHAPES (2D LOGIC)		
ID	Topic	What it tests
1	**Similarities and Like Shapes**	Find the shapes in a group that share a hidden common rule.
2	**Odd One Out**	Identify the one figure that breaks the established pattern.
3	**Analogies**	Shape A is to Shape B as Shape C is to... Identify the change and apply it.
4	**Series and Sequences**	Find the next shape in a row of five evolving boxes.
5	**Matrices (2x2 and 3x3)**	Solve grid puzzles by identifying the rule governing each row and column.
6	**Code Breakers**	Link shapes to letter codes based on their features (size, shading, type).

MANIPULATING SHAPES (2D AND 3D SPATIAL)		
ID	Topic	What it tests
7	**Rotations and Reflections**	Turn shapes by a given angle, or flip them as if held up to a mirror.
8	**Embedded Shapes**	Find a hidden target shape buried within a larger, more complex diagram.
9	**Following Folds**	Predict the pattern of holes or cuts after folded paper is opened out flat.
10	**Nets and Cubes**	Work out which 3D cube is formed when a flat net is folded up.
11	**3D Views**	Match a 3D block structure to its bird's-eye or side-on plan view.
12	**Block Counting**	Count the total number of blocks in a structure, including hidden ones.

Which topics appear most often?

Matrices, Series and Sequences, and Analogies account for the majority of marks in most GL and CEM papers. Nets and Cubes is the most commonly dropped topic because students do not practise visualising 3D space. Both areas deserve equal preparation time.

- Topics 3, 4, and 5 (Analogies, Sequences, Matrices) are the highest-frequency question types.
- Topics 9 and 10 (Folds and Nets) have the highest average mark loss. Practise these more.
- Topic 6 (Code Breakers) is highly learnable once the method is understood. See Section 04.

03 Non-Verbal Reasoning

General Mnemonics and Techniques to help with questions

7 min read · Section 03 of 05

Two systematic frameworks cover almost every NVR question type. Learn them in order. SNAPSS handles all questions involving movement and pattern. The NETS strategy handles cube and net questions.

The SNAPSS Checklist

Use SNAPSS whenever shapes are moving, changing, or evolving across a sequence. Run through every letter on every shape before looking at the answer options. This prevents the most common mark-losing mistake: spotting one change and choosing an answer before checking the rest.

S Shape	What kind of shape is it? Regular vs irregular. How many sides? Does it have curves?
N Number	Count everything: dots, stripes, lines inside the shape, or shapes within shapes.
A Angle	Has the shape been rotated? By how much: 45 degrees, 90 degrees, or 180 degrees?
P Position	Where is the shape sitting? Top, bottom, left, right, centre, or inside another shape?
S Size	Has the shape grown larger or shrunk smaller compared to the previous step?

S
Shading

Is it filled black, white, grey, striped, or dotted? Has the shading changed?

The NETS Strategy for Cubes

Apply these three rules in order when a question shows a flat net and asks which cube it forms. Each rule eliminates wrong answers quickly, so you arrive at the correct answer by a process of structured elimination rather than guesswork.

1

Eliminate the Duds

Cross out any cube that shows a face (shape or colour) that does not appear anywhere on the flat net. These are outright impossible and can be eliminated immediately without further checking.

2

Check the Opposites Rule

On a flat net, any two squares separated by exactly one square in between are on opposite faces of the finished cube. Opposite faces can never be seen at the same time. If an answer option shows two opposite faces both visible, eliminate it.

3

Check Anchor Point Orientation

Pick one face that has a clear direction (an arrow, a letter, an asymmetric pattern). Trace where it points on the net, then check whether it points in the same direction on the cube. If the orientation is wrong, eliminate that option.

Why systematic elimination beats visualisation

Most students try to mentally fold the net into a cube and then compare it to the options. This is unreliable and slow. The three-rule approach above eliminates wrong answers using simple logic. In most questions, two rules are enough to find the correct answer without ever needing to fully visualise the 3D object.

04 Non-Verbal Reasoning

Worked Question Walkthroughs

6 min read · Section 04 of 05

Two fully worked examples: one for Code Breakers (a logic-based topic) and one for Following Folds (a spatial topic). Both use only the systematic methods from Section 03. No guessing.

Walkthrough 1: Code Breakers

Question

A black circle is coded AX. A white circle is coded AY. A black square is coded BX. What is the code for a white square?

Working Through It

1. Isolate the first letter. Look at AX and AY. Both entries are circles, and both start with A. Therefore A means circle.
2. Isolate the second letter. Look at AX and BX. Both entries are black, and both end in X. Therefore X means black.
3. Deduce the remaining values. AY is a white circle, and we know A means circle, so Y must mean white. BX is a black square, and we know B is not circle (because A is circle), so B must mean square.
4. Build the answer. A white square is white (Y) and square (B). The code is BY.

Answer

The answer is BY.

Walkthrough 2: Following Folds

Question

A square piece of paper is folded in half vertically. A small triangle is then cut from the folded edge. What does the paper look like when it is unfolded?

Working Through It

1. Identify the fold line. A vertical fold creates a vertical line of symmetry down the centre of the paper.
2. Apply the Mirror Rule. Every fold line acts as a mirror. Whatever is cut on one side of the fold is automatically duplicated on the other side when the paper is opened.
3. Visualise the cut. The triangle is cut from the folded edge, which is the vertical centre. When unfolded, the two triangular cuts on either side of the fold meet to form a complete diamond shape.
4. Confirm the position. Because the cut was on the fold itself (the centre), the resulting hole appears in the exact centre of the paper.

Answer

The unfolded paper shows a diamond-shaped hole in the centre.

How to practise walkthroughs effectively

After working through any example question, always write out your reasoning in step form before checking the answer. If your answer is correct but your steps are missing or muddled, the method is not yet secure and will break down under exam pressure.

- Correct answer with correct steps: the method is secure.
- Correct answer with vague steps: you were lucky. Practise the steps.
- Wrong answer: identify exactly which step failed. Was it missing a change, misreading a code, or a flipped orientation?

05 Non-Verbal Reasoning

How to Structure Your Revision

5 min read · Section 05 of 05

A structured approach produces measurably better results than random practice. Follow these three steps in order, then apply the preparation tips to make the habit stick.

1. THE SCAN DRILL

Pick five Odd One Out questions from a practice paper. Before choosing any answer, run through the full SNAPSS checklist on every single shape in the group, even if you think you can see the answer after the first shape.

The purpose is not to get the right answer quickly. It is to build the automatic habit of checking all six attributes before deciding. Speed will come naturally once the checklist becomes instinctive. This is called building Visual Stamina.

2. THE ERROR AUDIT

After every practice session, go back through every question you got wrong and label the type of mistake.

- ***Subtle Difference:*** You saw the right type of change but missed a fine detail: a slightly different rotation, one extra line, or a small difference in shading density.
- ***Multiple Change:*** The shape was changing in two ways at once (rotating AND changing shading, for example) and you only tracked one of the changes.
- ***Wrong Topic:*** You looked for the wrong thing because you misread the question type. This is fixed by reading the instructions twice before starting.

Labelling mistakes this way tells you exactly which part of the SNAPSS checklist to focus on in your next session.

3. BUILD IT TO UNDERSTAND IT

If spatial questions (Nets, Folds, 3D Views, Block Counting) are not clicking from looking at diagrams alone, make them physical.

- Use LEGO bricks to build the 3D structures from Block Counting questions and count the hidden layers yourself.
- Print out a net from a practice paper, cut it out, and physically fold it into a cube. Do this three or four times and the Opposites Rule becomes immediately obvious.
- For Rotations, draw the shape on a piece of paper and physically rotate the paper by 90 or 180 degrees. Your eye will confirm what your brain struggles to picture.

Top Preparation Tips

Read the question before looking at the shapes

Looking for the Odd One Out when the question asks for Next in Sequence is one of the most common mark-losers in NVR. The question type changes what you are looking for entirely. Read the instruction first, every time.

The Skip-and-Return method

If the rule does not become clear within 45 seconds, make an educated guess, mark the question lightly with a small dot in the margin, and move on. Collect all the easy marks first. Return to the marked questions in the time remaining at the end.

Isolate one change at a time

If a shape is rotating AND changing shading simultaneously, ignore the shading completely in your first pass. Eliminate all options that are in the wrong position or rotation. Then check the shading only among the remaining options. Trying to track two changes at once leads to errors.

Draw it out on scrap paper

If you cannot mentally picture a rotation, draw the shape on your scrap paper and physically turn the page. If you cannot picture a fold, sketch the fold line and mark the cut. Drawing is not wasted time. Visualising incorrectly and choosing the wrong answer is wasted time.